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Abstract:

This document defines data types, functions and other basic components of the PKCS #11 Cryptoki interface.

Status:

This document was last revised or approved by the OASIS PKCS 11 TC on the above date. The level of approval is also listed above. Check the "Latest stage" location noted above for possible later revisions of this document. Any other numbered Versions and other technical work produced by the Technical Committee (TC) are listed at https://www.oasis-open.org/committees/tc_home.php?wg_abbrev=pkcs11#technical.

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The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC 2119] and [RFC 8174] when, and only when, they appear in all capitals, as shown here.

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1 Introduction

This document intends to meet this OASIS requirement on conformance clauses for providers and consumers of cryptographic services via PKCS#11 ([PKCS11_Spec] Section 7 - PKCS#11 Implementation Conformance) through profiles that define the use of PKCS#11 data types, objects, functions and mechanisms within specific contexts of provider and consumer interaction. These profiles define a set of normative constraints for employing PKCS#11 within a particular environment or context of use. They may, optionally, require the use of specific PKCS#11 functionality or in other respects define the processing rules to be followed by profile actors.

For normative definition of the elements of PKCS#11 specified in these profiles, see the PKCS#11 Specification [PKCS11_Spec].

2 Profiles

This document defines a selected set of conformance clauses which form PKCS #11 Profiles. A profile may be standalone or may be specified in terms of changes relative to another profile.

The PKCS 11 TC also welcomes proposals for new profiles. PKCS 11 TC members are encouraged to submit these proposals to the PKCS 11 TC for consideration for inclusion in a future version of this TC-approved document.

2.1 Profile Requirements

The following items SHALL be addressed by each profile:

1. Specify the versions of the PKCS#11 specification that SHALL be supported if versions other than [PKCS11_Spec] are supported
2. Specify the list of additional data types that SHALL be supported
3. Specify the list of additional attributes that SHALL be supported
4. Specify the list of additional objects that SHALL be supported
5. Specify the list of additional functions that SHALL be supported
6. Specify the list of additional mechanisms that SHALL be supported
7. Specify any other requirements that SHALL be supported
8. Specify any mandatory test cases that SHALL be supported by conforming implementations
9. Specify optional test cases that MAY be supported by conforming implementations

Note: items may be specified either directly in a profile or by reference to other profiles. Where another profile is referenced as required, the combination of the requirements of all referenced required profiles (directly or indirectly) SHALL apply.

2.2 Guidelines for other Profiles

Any vendor or organization, such as other standards bodies, MAY create a PKCS#11 Profile and publish it.

1. The profile SHALL be publicly available.
2. The PKCS11 Technical Committee SHALL be formally advised of the availability of the profile and the location of the published profile.
3. The profile SHALL meet all the requirements of section 2.1
4. The PKCS11 Technical Committee SHOULD review the profile prior to final publication.

2.3 Defined Profile Identifiers

Profile objects (object class *CKO_PROFILE*) describe which PKCS #11 profiles a provider implements.

The *CKA_PROFILE_ID* attribute identifies a profile that the provider implements.

Attributes	Data Types	Meaning
CKA_PROFILE_ID	CK_PROFILE_ID	ID of the supported profile

The following table defines the CK_PROFILE_ID values:

Constant	Meaning
CKP_INVALID_ID	Invalid Profile
CKP_BASELINE_PROVIDER	Baseline Provider
CKP_EXTENDED_PROVIDER	Extended Provider

CKP_AUTHENTICATION_TOKEN	Authentication Token Provider or Consumer
CKP_PUBLIC_CERTIFICATES_TOKEN	Public Certificates Token Provider or Consumer
CKP_COMPLETE_PROVIDER	Complete Provider
CKP_HKDF_TLS_TOKEN	HKDF TLS Token
CKP_VENDOR_DEFINED	Vendor defined

3 Conformance Test Cases

The test cases define a sequence of PKCS#11 function calls with specified input and output parameters. Each test case is provided in the XML format specified in PKCS#11 XML Representation (4) intended to be both human-readable and usable by automated tools.

Each test case has a unique label (the section name) which includes indication of mandatory (-M-) or optional (-O-) status and the specification version major and minor numbers as part of the identifier.

The test cases may depend on a specific configuration of a PKCS#11 provider and consumer and being configured in a manner consistent with the test case assumptions.

Where possible the flow of identifiers between tests, date values, and other dynamic items are indicated using symbolic identifiers – in actual request and response messages these dynamic values will be filled in with valid values.

Symbolic identifiers SHALL be of the form `${ParameterName}`. Wherever a symbolic identifier occurs in a test case the implementation must replace it with a reasonable appearing datum of the expected type.

The symbolic identifier may reference return parameters or array or list items by index number. Array index numbers SHALL be of the form `${ParameterName[ArrayIndex]}` and the first element SHALL be indicated by index zero.

The symbolic identifier may reference elements nested within other elements. Nested references SHALL be of the form `${ParameterName.SubElement}` and MAY also include an array index.

Note: the values for the returned items are illustrative. Actual values from a real consumer or provider MAY vary as specified in section 3.1.

3.1 Permitted Test Case Variations

Whilst the test cases provided in a Profile define the allowed call and return content, some inherent variations MAY occur and are permitted within a successfully completed test case.

Each test case MAY include allowed variations in the description of the test case in addition to the variations noted in this section.

Other variations not explicitly noted in this section SHALL be deemed non-conformant.

3.1.1 Variable Items

An implementation conformant to a Profile MAY vary the following values (expressed using the XML name for the items):

Provider specific information within the Info, SlotInfo and TokenInfo elements:

1. LibraryDescription
2. LibraryVersion
3. ManufacturerID
4. SlotDescription
5. HardwareVersion
6. FirmwareVersion
7. serialNumber
8. label
9. model
10. utcTime

Session specific information:

1. SlotID
2. Object

230 3. Session
231 Object specific information:
232 1. Object
233 Operation specific information:
234 1. Data
235 2. EncryptedData
236 3. RandomData
237 Attribute specific information:
238 1. VALUE
239 2. PUBLIC_EXPONENT
240 3. PRIVATE_EXPONENT
241 4. PRIME_1
242 5. PRIME_2
243 6. EXPONENT_1
244 7. EXPONENT_2
245 8. COEFFICIENT
246 9. PRIME
247 10. SUBPRIME
248 11. BASE
249 12. EC_POINT
250 13. UNIQUE_ID

251 **3.1.2 Variable behavior**

252 An implementation conformant to a Profile SHALL allow variation of the following behavior:
253 1. A test may omit the clean-up functions at the end of the test provided there is a separate
254 mechanism to remove the created objects during testing.
255 2. A test may omit the test identifiers in various attributes if the consumer is unable to include them
256 in calls.
257 3. The number of entries and order of entries in the list returned in the *C_GetSlotList*,
258 *C_GetMechanismList*, and *C_GetInterfaceList* functions make vary, provided that at least one
259 entry within the list matches the logical context of the test case.
260

4 PKCS#11 XML Representation

4.1 Normalizing Names

PKCS#11 parameter and structure field names SHALL be normalized to create a 'CamelCase' format that would be suitable to be used as a variable name in C/Java or an XML element name.

Hungarian notation type indicators are entirely omitted from names (i.e. *h*, *ph*, *ul*, *pul*, and *p* are omitted).

PKCS#11 function names are represented as-is (unchanged) as XML elements of the same name.

4.2 Omitted Items

PKCS#11 pointers for callback functions and reserved items are entirely omitted (i.e. *pApplication*, *pReserved*, *Notify* are not present).

Hungarian notation type indicators are entirely omitted from names (i.e. *h*, *ph*, *ul*, *pul*, and *p* are omitted).

4.3 Value Representation

The value for PKCS#11 binary (*CK_BYTE*) information SHALL be encoded as hexadecimal strings.

The value for PKCS#11 textual information (*CK_CHAR*, *CK_UTF8CHAR*) SHALL be encoded as hex strings.

The value for PKCS#11 numeric information SHALL be encoded as integers or as hexadecimal strings.

4.3.1 Enumerated Type Representation

Each PKCS#11 type value SHALL be represented in string/text form using the uppercase C macro name with the type prefix omitted. E.g. *CKR_OK* has a representation of "OK".

4.3.2 Boolean Representation

Each PKCS#11 boolean value (*CK_BBOOL*) SHALL be represented in string/text form either as "true" (non-zero) or "false" (zero). No other representation SHALL be used.

4.3.3 Flag Type Representation

Each PKCS#11 flag value SHALL be represented using the uppercase C macro names with the type prefix omitted for each bit. If multiple bit flags are set then each SHALL be present separated by either a space (' ') or a pipe ('|') character.

4.3.4 Special Value Representation

For PKCS#11 *CK_ULONG* values which have special interpretation as *CK_UNAVAILABLE_INFORMATION* or *CK_EFFECTIVELY_INFINITE* the string values "UnavailableInformation" and "EffectivelyInfinite" SHOULD be used instead of the numeric values to improve readability. This approach is used in the *CK_TOKEN_INFO* structure for various count and length and size values.

4.3.5 Function Call and Return Representation

PKCS#11 function calls are represented as an XML element of the same name containing the input parameters each represented as XML elements and an XML element of the same name as the PKCS#11 function name with an XML element attribute named *rv* containing the return value. The XML element for the input parameters is always immediately followed by the XML element for the output results.

297 PKCS#11 parameters and structure members that are not arrays or lists are represented as XML
298 elements with the value of the parameter or structure member contained within the XML element attribute
299 *value*.

300 4.3.6 Array and List Representation

301 PKCS#11 parameters and structure members that are arrays or lists are represented as XML elements
302 with the length of the array or list contained in XML element attribute *length* and the members of the array
303 or list represented as nested XML elements unless an XML element attribute-based representation has
304 been separately defined (e.g for *CK_ATTRIBUTE*).

305 PKCS#11 parameters and structure member elements that represent the count of arrays are omitted as
306 input parameters as the lengths can be determined by a count of the number of XML elements within the
307 call or return XML element within the element representing the PKCS#11 function call.

308 4.3.7 Determining Array or List Length

309 The PKCS#11 approach of passing in a NULL pointer value and using an input/output parameter to
310 determine the required pointer buffer length for a subsequent call SHALL be encoded as request where
311 the XML element for pointer has no specified value or length for the function call and the returned length
312 is contained in the XML element attribute *length*.

313 4.3.8 Hexadecimal String Encoding

314 Hexadecimal strings SHALL NOT include any white space.

315 Hexadecimal strings SHALL use either uppercase 'A'-'F' or lowercase 'a'-'f' along with '0' to '9'.

316 Numeric values represented as hexadecimal strings SHALL begin with '0x'.

317 Binary values represented as hexadecimal strings SHOULD omit the '0x'.

318 4.4 XML Root Element

319 XML documents representing a sequence of PKCS#11 function calls and returns SHALL have an XML
320 root element of *PKCS11*.

321 4.5 XML Namespaces

322 If namespaces are necessary within a specific context, then each XML element SHALL use the following
323 namespace:

324 urn:oasis:tc:pkcs11:xm1ns

325 4.6 XML Element Encoding

326 For XML, each function call is represented as a sequence of two XML element with optional attributes.

327 The parameters to each call are represented as nested XML elements, and any structures used within
328 those parameters are represented as nested XML elements within the nested XML elements.

329 The types of each parameter or structure element are fixed within the PKCS#11 specification and are not
330 separately represented within the XML encoding. i.e. the types are inherently known by implementations
331 and are fixed, matching the underlying C static type declaration.

332 4.6.1 Boolean

333 XML value uses [XML-SCHEMA] type xsd:Boolean. The value SHALL be FALSE, false, TRUE or true.

334 `<TokenPresent value="false"/>`

4.6.2 Text String

XML value uses [XML-SCHEMA] type xsd:string

```
<Pin value="12345678"/>
```

4.6.3 Byte String

XML value uses [XML-SCHEMA] type xsd:hexBinary

```
<EncryptedData value="8dce78ad"/>
```

4.6.4 Enumerated Type

XML value uses [XML-SCHEMA] type xsd:string and is either a hexadecimal string or the *Enumerated Type Representation* name. If an XSD with xsd:enumeration restriction is used to define valid values parsers should also accept any hexadecimal string in addition to the defined enumeration values to allow for user extensions and non-textual encoding parsers.

```
<Type value="AES_CBC"/>  
<Type value="0x00001082"/>  
<Type value="4426"/>
```

4.6.5 Function Call and Return

PKCS#11 function call and return SHALL be encoded as an XML element for the function call with any required parameters as nested XML elements, followed by an XML element for the function return with an XML element attribute of *rv* containing the return code from the function call encoded as an Enumerated Type and any output parameters as nested XML elements.

```
<C_Initialize/>  
<C_Initialize rv="OK"/>  
<C_GetSlotList>  
  <TokenPresent value="false"/>  
  <SlotList/>  
</C_GetSlotList>  
<C_GetSlotList rv="OK">  
  <SlotList length="1"/>  
</C_GetSlotList>
```

4.6.6 Attribute

PKCS#11 attributes (*CK_ATTRIBUTE*) SHALL be encoded as an XML element with an XML element attribute *type* containing the name of the PKCS#11 attribute and an XML element attribute *value* containing the value of the attribute. Where the PKCS#11 attribute has a specified type, the *value* SHALL be encoding using the encoding rules for that type of PKCS#11 value.

```
<Attribute type="CLASS" value="SECRET_KEY"/>  
<Attribute type="KEY_TYPE" value="AES"/>  
<Attribute type="LABEL" value="timing-key"/>  
<Attribute type="TOKEN" value="TRUE"/>  
<Attribute type="PRIVATE" value="TRUE"/>  
<Attribute type="EXTRACTABLE" value="TRUE"/>  
<Attribute type="SENSITIVE" value="TRUE"/>  
<Attribute type="ENCRYPT" value="TRUE"/>  
<Attribute type="DECRYPT" value="TRUE"/>  
<Attribute type="VALUE_LEN" value="16"/>
```

380 **4.6.7 XML Element Attributes**

381 XML element attributes other than “type”, “value”, “length” and “rv” as defined in this specification SHALL
382 not be used. All other PKCS#11 concepts are represented as XML elements and not XML element
383 attributes.
384

5 Base Profiles

The following subsections describe currently-defined profiles related to the use of PKCS #11. The profiles define classes of PKCS #11 functionality to which an implementation can declare conformance.

5.1 Baseline Provider

A PKCS #11 provider makes cryptographic functionality available to a consuming application in terms of the PKCS #11 API.

This profile specifies the most basic functionality that would be expected of a conformant PKCS #11 provider – the ability to provide information about the capabilities of the cryptographic services provided.

An implementation conforms to this specification as a Baseline Provider if it meets the following conditions:

1. Supports the conditions required by the *PKCS#11 Provider Implementation Conformance* clauses [PKCS11_Spec]
2. Supports the following data types [PKCS11_Spec]
 - a. *CK_VERSION*
 - b. *CK_INFO*
 - c. *CK_SLOT_ID*
 - d. *CK_SLOT_INFO*
 - e. *CK_TOKEN_INFO*
 - f. *CK_SESSION_HANDLE*
 - g. *CK_USER_TYPE*
 - h. *CK_SESSION_INFO*
 - i. *CK_OBJECT_HANDLE*
 - j. *CK_OBJECT_CLASS*
 - k. *CK_ATTRIBUTE_TYPE*
 - l. *CK_ATTRIBUTE*
 - m. *CK_PROFILE_ID*
 - n. *CK_RV*
 - o. *CK_FUNCTION_LIST*
 - p. *CK_INTERFACE*
 - q. *CK_C_INITIALIZE_ARGS*
3. Supports the following attributes [PKCS11_Spec]:
 - a. *CKA_CLASS*
 - b. *CKA_TOKEN*
 - c. *CKA_VALUE*
 - d. *CKA_ID*
 - e. *CKA_PRIVATE*
 - f. *CKA_MODIFIABLE*
 - g. *CKA_LABEL*
 - h. *CKA_UNIQUE_IDENTIFIER*
 - i. *CKA_PROFILE_ID*
4. Supports the following objects [PKCS11_Spec]:
 - a. *CKO_PROFILE* with value *CKP_BASELINE_PROVIDER*
5. Supports the following functions [PKCS11_Spec]:
 - a. *C_GetFunctionList*
 - b. *C_GetInterfaceList*
 - c. *C_GetInterface*
 - d. *C_Initialize*
 - e. *C_Finalize*
 - f. *C_GetInfo*
 - g. *C_GetSlotList*
 - h. *C_GetSlotInfo*
 - i. *C_GetTokenInfo*

- 437 j. *C_OpenSession*
- 438 k. *C_CloseSession*
- 439 l. *C_GetSessionInfo*
- 440 m. *C_FindObjectsInit*
- 441 n. *C_FindObjects*
- 442 o. *C_FindObjectsFinal*
- 443 p. *C_GetAttributeValue*
- 444 6. Supports the following mechanisms:
- 445 a. None specified
- 446 7. Supports *Error Handling* [PKCS11_Spec] for any supported object, function or mechanism
- 447 8. Optionally supports any clause within [PKCS11_Spec] that is not listed above
- 448 9. Optionally supports extensions outside the scope of this standard (e.g., vendor defined
- 449 extensions, conformance clauses) that do not contradict any PKCS #11 requirements

450 5.1.1 Baseline Provider Mandatory Test Cases

451 5.1.1.1 BL-M-1-32

452 See <test-cases/pkcs11-v3.2/mandatory/BL-M-1-32.xml>

453 5.2 Complete Provider

454 A PKCS #11 provider makes cryptographic functionality available to a consuming application in terms of
455 the PKCS #11 API.

456 This profile specifies the functionality that would be expected of a conformant PKCS #11 provider that
457 implements the entire specification.

458 An implementation conforms to this specification as a Complete Provider if it meets the following
459 conditions:

- 460 1. Supports the conditions required by the *PKCS#11 Provider Implementation Conformance*
- 461 clauses [PKCS11_Spec]
- 462 2. Supports all data types [PKCS11_Spec]
- 463 3. Supports all attributes [PKCS11_Spec]
- 464 4. Supports all objects [PKCS11_Spec]
- 465 5. Supports all functions [PKCS11_Spec]
- 466 6. Supports all mechanisms [PKCS11_Spec] Section 6
- 467 7. Supports *Error Handling* [PKCS11_Spec]
- 468 8. Optionally supports extensions outside the scope of this standard (e.g., vendor defined
- 469 extensions, conformance clauses) that do not contradict any PKCS #11 requirements

470 5.3 Extended Provider

471 This profile builds on the PKCS#11 Baseline Provider to add support for mechanism-based usage.

472 An implementation conforms to this specification as an Extended Provider if it meets the following
473 conditions:

- 474 1. Supports the conditions required by the PKCS #11 conformance clauses ([PKCS11_Spec]
- 475 Section 7 (PKCS#11 Implementation Conformance)
- 476 2. Supports the conditions required by the PKCS #11 Baseline Provider clauses section 5.1.
- 477 3. Supports the following data types [PKCS11_Spec]:
- 478 a. *CK_MECHANISM_TYPE*
- 479 b. *CK_MECHANISM*
- 480 4. Supports the following attributes [PKCS11_Spec]:
- 481 a. None specified
- 482 5. Supports the following objects [PKCS11_Spec]:
- 483 a. *CKO_PROFILE* with value *CKP_EXTENDED_PROVIDER*
- 484 6. Supports the following functions [PKCS11_Spec]:

- 485 a. C_GetMechanismList
- 486 b. C_GetMechanismInfo
- 487 c. C_Login
- 488 d. C_LoginUser
- 489 e. C_Logout
- 490 7. Supports the following mechanisms:
- 491 a. None specified
- 492 8. Supports *Error Handling* [PKCS11_Spec] for any supported object, function or mechanism
- 493 9. Optionally supports any clause within [PKCS11_Spec] that is not listed above
- 494 10. Optionally supports extensions outside the scope of this standard (e.g., vendor defined
- 495 extensions, conformance clauses) that do not contradict any PKCS #11 requirements

496 5.3.1 Extended Provider Mandatory Test Cases

497 5.3.1.1 EXT-M-1-32

498 See <test-cases/pkcs11-v3.2/mandatory/EXT-M-1-32.xml>

499 5.4 Authentication Token

500 This profile builds on the PKCS #11 Baseline Provider and/or Baseline Consumer profiles to provide for
501 use in the context of an authentication token.

502 An implementation conforms to this specification as an Authentication Token if it meets the following
503 conditions:

- 504 1. If the implementation is a consumer then it SHALL support the conditions required by the
- 505 PKCS #11 Baseline Consumer Clause (Section 5.7)
- 506 2. If the implementation is a provider then it SHALL support the conditions required by the
- 507 PKCS #11 Baseline Provider Clause (Section 5.1)
- 508 3. Supports the following data types [PKCS11_Spec]:
- 509 a. None specified
- 510 4. Supports the following attributes [PKCS11_Spec]:
- 511 a. None specified
- 512 5. Supports the following objects [PKCS11_Spec]:
- 513 a. CKO_PRIVATE_KEY
- 514 b. CKO_PUBLIC_KEY
- 515 c. CKO_PROFILE with value CKP_AUTHENTICATION_TOKEN
- 516 6. Supports the following functions [PKCS11_Spec]:
- 517 a. C_Login
- 518 b. C_LoginUser
- 519 c. C_Logout
- 520 d. C_SignInit
- 521 e. C_Sign and/or C_SignUpdate and C_SignFinal
- 522 7. Supports the following mechanisms:
- 523 a. None specified
- 524 8. Supports *Error Handling* [PKCS11_Spec] for any supported object, function or mechanism
- 525 9. Optionally supports any clause within [PKCS11_Spec] that is not listed above
- 526 10. Optionally supports extensions outside the scope of this standard (e.g., vendor defined
- 527 extensions, conformance clauses) that do not contradict any PKCS #11 requirements.

528 5.4.1 Authentication Token Provider Mandatory Test Cases

529 5.4.1.1 AUTH-M-1-32

530 See <test-cases/pkcs11-v3.2/mandatory/AUTH-M-1-32.xml>

531

5.5 Public Certificates Token

This profile builds on the PKCS #11 Baseline Provider and/or Baseline Consumer profiles to provide for use in the context of a public certificates token.

An implementation conforms to this specification as a Public Certificates Token if it meets the following conditions:

1. If the implementation is a consumer then it SHALL support the conditions required by the PKCS #11 Baseline Consumer Clause (Section 5.7)
2. If the implementation is a provider then it SHALL support the conditions required by the PKCS #11 Baseline Provider Clause (Section 5.1)
3. Supports the following data types [PKCS11_Spec]:
 - a. None specified
4. Supports the following attributes [PKCS11_Spec]:
 - a. None specified
5. Supports the following objects [PKCS11_Spec]:
 - a. *CKO_CERTIFICATE*
 - b. *CKO_PROFILE* with value *CKP_PUBLIC_CERTIFICATES_TOKEN*
6. Supports the following functions [PKCS11_Spec]:
 - a. None specified
7. Supports the following mechanisms [PKCS11_Spec]:
 - a. None specified
8. Supports the following object location requirements:
 - a. All certificates are publicly readable, able to be found on the token without a login having been performed
 - b. All certificates for which a matching private key also exists on the token must have a matching *CKA_ID* attribute for the certificate and private key
 - c. One or more of the following conditions must be met:
 - i. The matching private key for a certificate can be found via *C_FindObjects* using the matching *CKA_ID* value without a login having been performed;
 - ii. The matching public key for a certificate can be found via *C_FindObjects* using the matching *CKA_ID* value without a login having been performed
9. Supports Error Handling [PKCS11_Spec] for any supported object, function or mechanism
10. Optionally supports any clause within [PKCS11_Spec] that is not listed above
11. Optionally supports extensions outside the scope of this standard (e.g., vendor defined extensions, conformance clauses) that do not contradict any PKCS #11 requirements.

5.5.1 Public Certificates Token Provider Mandatory Test Cases

5.5.1.1 CERT-M-1-32

See <test-cases/pkcs11-v3.2/mandatory/CERT-M-1-32.xml>

5.6 HKDF TLS Token

This profile builds on the PKCS #11 Baseline Provider and/or Baseline Consumer profiles to provide for use in the context of TLS 1.3 connections using the CKM_HKDF_DERIVE_DATA mechanism.

An implementation conforms to this specification as an HKDF TLS Token if it meets the following conditions:

1. If the implementation is a consumer then it SHALL support the conditions required by the PKCS #11 Baseline Consumer Clause (Section 5.7)
2. If the implementation is a provider then it SHALL support the conditions required by the PKCS #11 Baseline Provider Clause (Section 5.1)
3. Supports the following data types [PKCS11_Spec]:
 - b. *CK_HKDF_PARAMS*
4. Supports the following attributes [PKCS11_Spec]:

- 581 a. None specified
- 582 5. Supports the following objects [PKCS11_Spec]:
- 583 a. *CKO_DATA*
- 584 b. *CKO_SECRET_KEY*
- 585 c. *CKO_PROFILE* with value *CKP_HKDF_TLS_TOKEN*
- 586 6. Supports the following functions [PKCS11_Spec]:
- 587 a. *C_DeriveKey*
- 588 7. Supports the following mechanisms:
- 589 a. *CKM_HKDF_DATA*
- 590 A conformant provider SHALL not reject derive requests based on the plnfo
- 591 value if the following plnfo values are given:
- 592 i. The string L1,L2,"tls iv",0 (L1, L2, 0x74, 0x6c, 0x73, 0x20, 0x69, 0x76, 0x00)
- 593 where L1 is the most significant byte of *CKA_VALUE_LEN* and L2 is the least
- 594 significant byte of *CKA_VALUE_LEN*.
- 595 ii. The string L1,L2,"tls quic iv",0 (L1, L2, 0x74, 0x6c, 0x73, 0x20, 0x71, 0x75,
- 596 0x69, 0x63, 0x20, 0x69, 0x76, 0x00) where L1 is the most significant byte of
- 597 *CKA_VALUE_LEN* and L2 is the least significant byte of *CKA_VALUE_LEN*.
- 598 A conformant provider MAY accept other values for plnfo.
- 599 8. Supports *Error Handling* [PKCS11_Spec] for any supported object, function or mechanism
- 600 9. Optionally supports any clause within [PKCS11_Spec] that is not listed above
- 601 10. Optionally supports extensions outside the scope of this standard (e.g., vendor defined
- 602 extensions, conformance clauses) that do not contradict any PKCS #11 requirements.
- 603

604 5.7 Baseline Consumer

605 A PKCS #11 consumer calls a PKCS #11 provider implementation of the PKCS #11 API in order to use
606 the cryptographic functionality from that provider.

607 This profile specifies the most basic functionality that would be expected of a conformant PKCS #11
608 consumer – the ability to consume information via the cryptographic services offered by a provider.

609 An implementation conforms to this specification as a Baseline Consumer if it meets the following
610 conditions:

- 611 1. Supports the conditions required by the *PKCS#11 Consumer Implementation Conformance*
- 612 clauses [PKCS11_Spec]
- 613 2. Supports the following data types [PKCS11_Spec]:
- 614 a. *CK_VERSION*
- 615 b. *CK_INFO*
- 616 c. *CK_SLOT_ID*
- 617 d. *CK_SLOT_INFO*
- 618 e. *CK_TOKEN_INFO*
- 619 f. *CK_SESSION_HANDLE*
- 620 g. *CK_USER_TYPE*
- 621 h. *CK_SESSION_INFO*
- 622 i. *CK_OBJECT_HANDLE*
- 623 j. *CK_OBJECT_CLASS*
- 624 k. *CK_ATTRIBUTE_TYPE*
- 625 l. *CK_ATTRIBUTE*
- 626 m. *CK_RV*
- 627 n. *CK_FUNCTION_LIST*
- 628 o. *CK_C_INITIALIZE_ARGS*
- 629 p. *CK_INTERFACE* (if *C_GetInterfaceList* and *C_GetInterface* is supported)
- 630 3. Supports the following attributes [PKCS11_Spec]:
- 631 a. *CKA_CLASS*

- 632 b. *CKA_VALUE*
633 4. Supports the following objects:
634 a. None specified
635 5. Supports the following functions [PKCS11_Spec]:
636 a. *C_GetFunctionList* or *C_GetInterfaceList* and *C_GetInterface*
637 b. *C_Initialize*
638 c. *C_Finalize*
639 d. *C_GetInfo*
640 e. *C_GetSlotList*
641 f. *C_GetSlotInfo*
642 g. *C_GetTokenInfo*
643 h. *C_OpenSession*
644 i. *C_CloseSession*
645 6. Supports the following mechanisms:
646 a. None specified
647 7. Supports *Error Handling* [PKCS11_Spec] for any supported object, function or mechanism
648 8. Optionally supports any clause within [PKCS11_Spec] that is not listed above
649 9. Optionally supports extensions outside the scope of this standard (e.g., vendor defined
650 extensions, conformance clauses) that do not contradict any PKCS #11 requirements

651 5.8 Extended Consumer

652 This profile builds on the PKCS#11 Baseline Consumer profile to add support for mechanism-based
653 usage.

654 An implementation conforms to this specification as an Extended Consumer if it meets the following
655 conditions:

- 656 1. Supports the conditions required by the PKCS11 conformance clauses ([PKCS11_Spec]
657 Section 7 (PKCS#11 Implementation Conformance))
658 2. Supports the conditions required by the PKCS11 Baseline Consumer clauses section 5.7
659 3. Supports the following data types [PKCS11_Spec]:
660 a. *CK_MECHANISM_TYPE*
661 b. *CK_MECHANISM*
662 4. Supports the following attributes [PKCS11_Spec]:
663 a. None specified
664 5. Supports the following objects [PKCS11_Spec]:
665 a. None specified
666 6. Supports the following functions [PKCS11_Spec]:
667 a. *C_GetMechanismList*
668 b. *C_GetMechanismInfo*
669 7. Supports the following mechanisms:
670 a. None specified
671 8. Supports *Error Handling* [PKCS11_Spec] for any supported object, function or mechanism
672 9. Optionally supports any clause within [PKCS11_Spec] that is not listed above
673 10. Optionally supports extensions outside the scope of this standard (e.g., vendor defined
674 extensions, conformance clauses) that do not contradict any PKCS #11 requirements

6 Conformance

The baseline provider and consumer profiles provide the most basic functionality that is expected of a conformant PKCS#11 consumer or provider. The complete provider profile defines a PKCS#11 provider that implements the entire specification. A PKCS#11 implementation conformant to this specification (the PKCS#11 Profiles) SHALL meet all the conditions documented in one or more of the following sections.

6.1 Baseline Provider Profile Conformance

PKCS#11 provider implementations conformant to this profile:

1. SHALL support [PKCS11_Spec];
2. SHALL support the Baseline Provider conditions (5.1) and;
3. SHALL support one or more of the Baseline Provider Mandatory Test Cases (5.1.1).

6.2 Complete Provider Profile Conformance

PKCS#11 provider implementations conformant to this profile:

1. SHALL support [PKCS11_Spec];
2. SHALL support the Complete Provider conditions (5.2) and;
3. SHALL support all of the provider conformance clauses contained within Conformance (6).

6.3 Extended Provider Profile Conformance

PKCS#11 provider implementations conformant to this profile:

1. SHALL support [PKCS11_Spec];
2. SHALL support the Extended Provider conditions (5.3) and;
3. SHALL support one or more of the Extended Provider Mandatory Test Cases (5.3.1).

6.4 Authentication Token Provider Profile Conformance

PKCS#11 provider implementations conformant to this profile:

1. SHALL support [PKCS11_Spec];
2. SHALL support the Authentication Token conditions (5.4) and;
3. SHALL support all of the Authentication Token Provider Mandatory Test Cases (5.4.1).

6.5 Public Certificates Token Provider Profile Conformance

PKCS#11 provider implementations conformant to this profile:

1. SHALL support [PKCS11_Spec];
2. SHALL support the Public Certificates Token conditions (5.5) and;
3. SHALL support all of the Public Certificates Token Provider Mandatory Test Cases (5.5.1).

6.6 HKDF TLS Token Provider Profile Conformance

PKCS#11 provider implementations conformant to this profile:

1. SHALL support [PKCS11_Spec];
2. SHALL support the HKDF TLS Token conditions (5.6).

709 **6.7 Baseline Consumer Profile Conformance**

710 PKCS#11 consumer implementations conformant to this profile:

- 711 1. SHALL support [PKCS11_Spec]; and
712 2. SHALL support the Baseline Consumer conditions (5.7).

713 **6.8 Authentication Token Consumer Profile Conformance**

714 PKCS#11 provider implementations conformant to this profile:

- 715 1. SHALL support [PKCS11_Spec]; and
716 2. SHALL support the Authentication Token conditions (5.4)

717 **6.9 Public Certificates Token Consumer Profile Conformance**

718 PKCS#11 provider implementations conformant to this profile:

- 719 1. SHALL support [PKCS11_Spec]; and
720 2. SHALL support the Public Certificates Token conditions (5.5)
721

7 PKCS #11 Implementation Conformance

7.1 PKCS#11 Consumer Implementation Conformance

An implementation is a conforming PKCS#11 Consumer if the implementation meets the conditions specified in one or more consumer profiles specified in [PKCS11-Prof].

A PKCS#11 consumer implementation SHALL be a conforming PKCS#11 Consumer.

If a PKCS#11 consumer implementation claims support for a particular consumer profile, then the implementation SHALL conform to all normative statements within the clauses specified for that profile and for any subclauses to each of those clauses.

7.2 PKCS#11 Provider Implementation Conformance

An implementation is a conforming PKCS#11 Provider if the implementation meets the conditions specified in one or more provider profiles specified in [PKCS11-Prof].

A PKCS#11 provider implementation SHALL be a conforming PKCS#11 Provider.

Appendix A. References

This appendix contains the normative and informative references that are used in this document. While any hyperlinks included in this appendix were valid at the time of publication, OASIS cannot guarantee their long-term validity.

A.1 Normative References

The following documents are referenced in such a way that some or all of their content constitutes requirements of this document.

[PKCS11_Spec]

PKCS #11 Specification Version 3.2. Edited by Dieter Bong and Greg Scott. Latest stage:
<https://docs.oasis-open.org/pkcs11/pkcs11-spec/v3.2/pkcs11-spec-v3.2.html>.

[RFC2119]

Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, DOI 10.17487/RFC2119, March 1997, <<https://www.rfc-editor.org/info/rfc2119>>.

[RFC8174]

Leiba, B., "Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words", BCP 14, RFC 8174, DOI 10.17487/RFC8174, May 2017, <<https://www.rfc-editor.org/info/rfc8174>>.

[XML]

Bray, Tim, et.al. eds, Extensible Markup Language (XML) 1.0 (Fifth Edition), W3C Recommendation 26 November 2008, <<http://www.w3.org/TR/2008/REC-xml-20081126>>.

A.2 Informative References

The following referenced documents are not required for the application of this document but may assist the reader with regard to a particular subject area.

[RFC3552]

Rescorla, E. and B. Korver, "Guidelines for Writing RFC Text on Security Considerations", BCP 72, RFC 3552, DOI 10.17487/RFC3552, July 2003, <<https://www.rfc-editor.org/info/rfc3552>>.

[XML-SCHEMA]

Paul V. Biron, Ashok Malhotra, XML Schema Part 2: Datatypes Second Edition, W3C Recommendation 26 November 2008, <<https://www.w3.org/TR/2004/REC-xmlschema-2-20041028>>.

Appendix B. Acknowledgments

B.1 Special Thanks

Substantial contributions to this document from the following individuals are gratefully acknowledged:

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Appendix C. Revision History

Revision	Date	Editor	Changes Made
WD01	23 Sep-2023	Tim Hudson	Initial Draft
WD02	16 Jul 2024	Tim Hudson	Updated XML encoding approach to avoid use of XML element attributes for structure fields to improve readability of the XML.
WD03	31 Jul 2024	Tim Hudson	Additional typos and cross reference errors and formatting corrected based on review feedback.
WD04	15 Apr 2025	Dieter Bong	Updated link to [PKCS11_Spec]; updated Appendix B Acknowledgements

Appendix D. Notices

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